

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 20

Code of Conduct

I Dinesh S(1925221152)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dinesh S_____

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Q1 : A cloud provider must secure thousands of virtual machines belonging to different customers. Recommend a security architecture that ensures tenant isolation.

Introduction

Cloud computing allows multiple customers (tenants) to share the same physical infrastructure while using separate virtual machines (VMs). Since resources such as servers, storage, and networks are shared, it is essential to ensure that one tenant cannot access another tenant's data or applications. A secure tenant isolation architecture protects customer information, prevents cyberattacks, and ensures privacy, availability, and compliance with security standards.

Recommended Security Architecture

1. Hypervisor-Based Isolation

- Use secure hypervisors such as VMware ESXi, Microsoft Hyper-V, KVM, or Xen.
- Each VM runs in an isolated environment with dedicated virtual CPU, memory, storage, and network resources.
- Regularly update and patch the hypervisor to prevent vulnerabilities and cross-VM attacks.

2. Network Segmentation

- Use Virtual LANs (VLANs), Virtual Private Clouds (VPCs), VXLANs, or Software-Defined Networking (SDN).
- Assign separate virtual networks to each tenant.
- Prevent communication between tenants unless explicitly permitted.

3. Identity and Access Management (IAM)

- Implement Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA) for administrators and users.
- Follow the Principle of Least Privilege (PoLP), allowing users access only to the resources required for their work.
- Maintain detailed logs of user activities.

4. Firewalls and Security Groups

- Configure firewall rules for every virtual machine.
- Allow only essential services and network ports.
- Block unauthorized inbound and outbound traffic.
- Use Network Security Groups (NSGs) to enforce access control policies.

5. Data Encryption

- Encrypt data at rest using disk encryption.
- Encrypt data in transit using SSL/TLS protocols.
- Protect encryption keys using a secure Key Management System (KMS).

6. Continuous Monitoring and Logging

- Monitor network traffic, login attempts, and system activities using SIEM tools.
- Detect suspicious behavior through intrusion detection and anomaly detection systems.
- Generate alerts for security incidents and unauthorized access attempts.

7. Patch and Vulnerability Management

- Regularly update hypervisors, operating systems, and applications.
- Perform vulnerability scanning and penetration testing.
- Apply security patches immediately to reduce exposure to known threats.

8. Backup and Disaster Recovery

- Schedule automatic backups of critical data.
- Store backups securely in different locations.
- Maintain disaster recovery plans to ensure quick restoration after failures or cyberattacks.

Advantages

- Provides complete isolation between tenants.
- Protects confidential customer data.
- Reduces the risk of cyberattacks and data leakage.
- Ensures regulatory compliance and improves customer trust.
- Enhances the availability, reliability, and security of cloud services.

Conclusion

A layered security architecture combining hypervisor isolation, network segmentation, IAM, firewalls, encryption, continuous monitoring, and regular patch management provides effective tenant isolation. This approach ensures that each customer's virtual machines remain secure, independent, and protected from unauthorized access in a multi-tenant cloud environment.

Q2: Evaluate the impact of open firewall ports on virtual machine security and suggest mitigation techniques.

Introduction

A firewall is one of the primary security mechanisms used to protect virtual machines by controlling network traffic. Firewall ports allow communication between a VM and external networks. However, leaving unnecessary ports open exposes the VM to cyber threats such as unauthorized access, malware infections, and denial-of-service attacks. Therefore, proper firewall configuration is essential for maintaining VM security.

Impact of Open Firewall Ports

1. Unauthorized Access

- Open ports may allow attackers to access services running on the virtual machine.
- Weak authentication or outdated services increase the risk of intrusion.

2. Increased Attack Surface

- Every open port becomes a potential entry point for attackers.
- More exposed services increase the likelihood of successful attacks.

3. Malware and Ransomware Attacks

- Vulnerable applications listening on open ports can be exploited to install malware or ransomware.
- Malware can compromise system files and steal sensitive information.

4. Data Breaches

- Attackers may gain access to confidential customer data stored on the VM.
- Data theft can result in financial losses and reputational damage.

5. Denial-of-Service (DoS/DDoS) Attacks

- Attackers may flood open ports with excessive traffic.
- This can overload system resources and make services unavailable.

6. Privilege Escalation

- Exploiting vulnerable services may allow attackers to obtain administrator privileges.
- Once elevated privileges are gained, attackers can fully control the VM.

7. Lateral Movement

- After compromising one VM, attackers may attempt to access other systems within the same cloud network if proper isolation is not implemented.

Mitigation Techniques

1. Close Unnecessary Ports

- Keep only essential ports open.
- Disable unused network services.

2. Configure Strong Firewall Rules

- Allow access only from trusted IP addresses.
- Block all unauthorized inbound and outbound traffic.

3. Use Network Security Groups (NSGs)

- Apply security policies to individual VMs and subnets.
- Restrict communication between different workloads.

4. Enable IDS/IPS

- Intrusion Detection Systems (IDS) monitor suspicious activities.
- Intrusion Prevention Systems (IPS) automatically block malicious traffic.

5. Regular Updates and Patch Management

- Keep operating systems, applications, and firewall software updated.
- Apply security patches as soon as they become available.

6. Strong Authentication

- Use strong passwords and Multi-Factor Authentication (MFA).
- Disable default accounts and unnecessary remote access services.

7. Continuous Monitoring

- Monitor firewall logs, user activities, and network traffic.
- Detect abnormal behavior early and respond quickly to security incidents.

Advantages of Proper Firewall Configuration

- Reduces the attack surface.
- Prevents unauthorized access.
- Protects sensitive data from cyber threats.
- Improves compliance with security policies.
- Ensures reliable and secure operation of virtual machines.

Conclusion

Open firewall ports can significantly weaken virtual machine security by increasing the risk of unauthorized access, malware infections, data breaches, and denial-of-service attacks. Organizations can minimize these risks by closing unnecessary ports, implementing strict firewall rules, enabling IDS/IPS, performing regular updates, using strong authentication, and continuously monitoring network traffic. These best practices create a secure cloud environment and protect virtual machines from evolving cyber threats.